FEATURE ENGINEERING & BASELINE REPORT

Project FORESIGHT — Stages 4, 5 & 6

Client: NorthBay Living | Zidio Development Internship | August 2026

1. Objective

This report documents the preparation of the weekly modelling dataset, the engineered features, and the seasonal-naive baseline. As required by the engagement brief, a simple baseline must be established before any complex model is attempted. Every later model must beat this baseline on WAPE.

2. Stage 5 — Weekly Dataset Preparation

Daily records were aggregated to weekly grain (SKU × week, week ending Sunday).

Item	Detail
Source	retail_forecasting_cleaned_v2 (365,000 daily rows)
Output grain	SKU × Week
Weekly rows	52,400 (200 SKUs × ~262 weeks)
Week range	2021-08-09 → 2026-08-10
Output file	weekly_forecast_ready.csv (26 columns)

3. Stage 4 — Feature Engineering

Features created for each SKU weekly series (no leakage — only past data used):

Feature	**Description**
lag_1, lag_2, lag_4	Demand from 1, 2 and 4 weeks earlier
lag_52	Same week last year (seasonal lag)
roll_mean_4 / roll_mean_8	Rolling mean of previous 4 / 8 weeks
roll_std_4	Rolling std (demand volatility)
weekofyear, month, quarter	Calendar features from week_start

4. Stage 6 — Seasonal-Naive Baseline

Forecast rule: **baseline_pred = lag_52** (same week last year). Fallback when lag_52 is missing: lag_4 → lag_1 → 0. This is the standard seasonal-naive method required by the brief.

5. Baseline Results (WAPE)

WAPE = Σ |Actual – Predicted| / Σ |Actual| × 100. Lower is better.

Metric	WAPE
Overall (full history)	**32.12%**
Last 1 year	**31.26%**
Decor	30.02%
Furniture	31.72%
Small Appliances	36.84%

Interpretation: The seasonal-naive baseline achieves ~31–32% WAPE. Any stronger model (LightGBM, etc.) must beat these numbers on a fair backtest. Small Appliances is the hardest category for the naive method.

6. Deliverables from this Stage

- weekly_forecast_ready.csv — 52,400 rows × 26 columns with features + baseline_pred
- Seasonal-naive baseline with documented WAPE scores
- This report

7. Next Steps

1. Train a stronger model (e.g. LightGBM) using the engineered features.
2. Evaluate with rolling-origin backtest and the same WAPE metric.
3. Keep the baseline if the complex model does not beat 31–32% WAPE; otherwise adopt the better model.
4. Proceed to risk scoring (stockout / overstock) using the chosen forecast.

8. Conclusion

Stages 4–6 are complete. A clean weekly modelling table, leakage-free features, and an honest seasonal-naive baseline (Overall WAPE 32.12%, Last-year WAPE 31.26%) are in place. The work is ready for model comparison in the next stage.